@page { margin: 14mm 12mm; } body { font-family: "DejaVu Sans", Arial, sans-serif; font-size: 10.5pt; line-height: 1.34; color: #1a2733; max-width: 100%; margin: 0; } h1 { font-size: 19pt; color: #12324b; border-bottom: 2px solid #12324b; padding-bottom: 3px; } h2 { font-size: 14pt; color: #12324b; border-left: 4px solid #12324b; padding-left: 7px; margin-top: 16px; } h3 { font-size: 12pt; color: #234; margin-top: 12px; } table { border-collapse: collapse; width: 100%; margin: 8px 0; font-size: 9pt; } th, td { border: 1px solid #b7c2cc; padding: 3px 6px; text-align: left; vertical-align: top; } th { background: #12324b; color: #fff; } tr:nth-child(even) td { background: #f2f5f7; } code { background: #eef1f4; padding: 1px 3px; border-radius: 3px; font-size: 8.8pt; } pre { background: #0f1b26; color: #dfe7ee; padding: 8px 10px; border-radius: 5px; font-size: 8.4pt; overflow-x: auto; white-space: pre-wrap; } pre code { background: transparent; color: inherit; }

ORAN_VIP_SUITE_GUIDE

O-RAN VIP Suite — Guide

Unified O-RAN VIP: 13 components covering 4G LTE (CPRI-over-Ethernet) and 5G NR (eCPRI Split 7.2x). Lane-1 (Verilator 5.020 + Z3) header/transaction-level VIP.

Layout

- common/pkg — shared eCPRI field model (pathfinder).
- components// — rtl/(pkg,codec), tb/, sim_main.cpp, spec.json, run.sh, docs/(GATE0,GATE1), sim_out/, gates/, corner/.
- gen/ — generator framework (single source of truth = SPECS in build_suite.py).
- regression/ — suite_regression*.json, gate4_8_report.json, code_cov.json.
- docs/ — FULL_COVERAGE_REPORT.md, COVERAGE_DASHBOARD.html, guide, architecture.
- signoff/ — ORAN_VIP_SUITE_SIGNED_OFF.md.

Run

python3 gen/build_suite.py # build+compile+GATE2/3 (coverage-enabled) RAND=100 python3 gen/run_gates.py # GATE4-8 (directed/random/negative/coverage) python3 gen/code_cov.py # GATE8 code coverage (stmt/branch + waivers) python3 gen/corner.py # corner/edge cases python3 gen/report.py # full regression + coverage reports + dashboard

Method (Lane-1)

Native randomize()+{} is silently ignored on Verilator 5.020 (measured), so the Z3 model IS the constraint block; every txn is proven-legal and seed-reproducible. A runtime legality self-check backstops the CONSTRAINTIGN hazard. Functional coverage via monitor-log COV-### engine; code coverage via custom sim_main coverage.dat. Reporter emits UVM-format summary; STATUS from executed evidence only (never inferred).

Adding a component

Add a spec entry to gen/build_suite.py SPECS (fields<=64b, legal, cross, cover) → rebuild → run.